

Arijit De

Kolkata, India

+918981800329 | arijitde2050@gmail.com | <https://github.com/arijitde92> | <https://www.linkedin.com/in/de-arijit/>

PROFESSIONAL SUMMARY

Machine Learning Engineer (PhD) with 5 years of experience owning backend architecture and delivery for production Python systems, including multi-agent/GenAI platforms serving 3,000+ agent runs/day. Skilled in Python, LangChain, LangGraph, FastAPI, asyncio, database schema design, and RAG pipelines over vector stores (AlloyDB), with hands-on experience in the Model Context Protocol (MCP). Comfortable owning end-to-end technical direction with architecture trade-offs, effort estimation, code review, mentoring, and presenting POC findings to technical and non-technical stakeholders.

WORK EXPERIENCE

AI / ML Engineer | Common Lab Ventures, Singapore | Remote Full time SEP 2025 – PRESENT

- Owned end-to-end architecture for a multi-agent AI platform (**LangGraph**, **Python**, **FastAPI**, **MCP**) powering Smart Project Management & Scheduling (SPMS), building a Postgres-checkpointed workflow graph with deterministic routing, retries, and human-approval interrupts.
- Built multi-modal **RAG** pipelines over text, image, and CAD datasets (10M+ vectors) on AlloyDB (pgvector + ScaNN) with row-level security and transactional ingestion, adaptive per-modality chunking, hybrid search, and cross-encoder reranking to achieve ~92% retrieval accuracy (recall@k) under ~600ms latency.
- Guided development and deployment of the platform on **GCP** (AlloyDB, Cloud Storage, Pub/Sub, OAuth2, RBAC) with 99% uptime, serving 100 concurrent users / 3,000 agent runs per day via an event-driven, continuous ingestion pipeline, increasing business efficiency by 60%.
- Hardened multi-agent workflows for production reliability with tool-validation layer (**Pydantic** schema + business-rule checks), agent guardrails, and human-in-the-loop approval gates for high-stakes actions cutting task failure rate by 60%.
- Instrumented observability via **LangSmith** tracing and **Ragas** evaluation, lifting evaluation scores by 75%.
- Translated real estate client requirements into technical specs, POC prototyping, work breakdown structures, and effort/risk estimates to keep releases on schedule.
- Led code reviews and mentored engineers on agentic system design and best practices, raising code quality and maintainability standards across the team.

AI / ML Engineer | mVizn, Singapore | Remote Full time JAN 2024 – AUG 2025

- Created a scalable end-to-end ML system from training to deployment for 3D semantic segmentation of indoor point clouds extracted from LiDAR scans using **PyTorch**, reducing Building Information Modeling (BIM) creation time by 60%.
- Used **MLflow** for model versioning and reproducible training workflows, cutting experimentation tracking time by 70%.
- Accomplished efficient data cleaning, data preprocessing and data ingestion pipelines using **Airflow** to enhance model training, data quality and data versioning by 75%.
- Designed relational databases using **PostgreSQL** enabling proper logging of user and model inference activities.
- Implemented scalable, robust backend architectures using RESTful API Design with **FastAPI**, data validation using **Pydantic**, asynchronous system design using asyncio, and multi-threaded programming, increasing concurrency by 40%.
- Improved reliable code development and reduced testing time by 40% by planning vigorous testing using **PyTest**.
- Organized weekly code review, mentoring sessions on deep learning model development to enhance team productivity.

Machine Learning Engineer | Mercedes-Benz Research & Development India, India | Onsite AUG 2018 – AUG 2019

- Developed and optimized a computer vision pipeline for Vulnerable Road User (VRU) detection by fine-tuning **YOLOv3** on large-scale annotated datasets, attaining detection mAP50 of 83% at real-time inference speeds; designed & deployed model training, evaluation, and validation workflows on **AWS (Sagemaker, EC2)**.
- Built RESTful APIs with **FastAPI**, tested via **Postman** for an interactive dashboard tracking data annotation and QA progress using **Python**, **Pandas** and **NumPy**, improving stakeholder visibility and trust.
- Built automated quality assurance (QA) and evaluation frameworks using Python, **OpenCV**, NumPy, improving annotation throughput by 20% and data quality by 25%.

Systems Engineer | Tata Consultancy Services, India | Onsite AUG 2014 – APR 2015

- Maintained high-availability backend systems (Java, MySQL) for TCS's internal website, serving 100,000+ users; streamlined the feature release cycle, reducing deployment time by 20%.

SKILLS

Languages:	Python, SQL
LLM/Agent Systems:	Multi-Agent Orchestration (LangGraph, Google ADK, CrewAI), LangChain, Retrieval Augmented Generation (RAG), Model Context Protocol (MCP), Agent Evaluation (Ragas, DeepEval), LLM Guardrails, Observability/Tracing (LangSmith Tracing), Agentic Memory, LLM APIs (OpenAI, Gemini, Claude)
Frameworks / Libraries:	FastAPI, Flask, Pydantic, asyncio, RESTful API, Postman, PyTest, MySQL, PyTorch, Scikit-learn, OpenCV, Open3D, Pandas, NumPy
Cloud & Ops:	GCP, AWS, Git, Docker, MLflow, GitHub
Actions Delivery & Leadership:	Code Review, Mentoring, Technical Architecture Ownership, Effort Estimation, Risk Assessment, Stakeholder & POC Presentations

EDUCATION

PhD Jadavpur University, India	FEB 2020 – APR 2025
M.Tech. Computer Science & Engineering Jadavpur University, India GPA - 8.83	AUG 2016 – JUN 2018
B.Tech. Computer Science & Engineering Techno India, India GPA - 8.81	AUG 2010 - MAY 2014

PROJECTS

- Doctor Appointment Scheduler | https://github.com/arijitde92/Doctor_Appointment_Scheduler
- Built a multi-agent conversational appointment-booking system (**Google ADK**) with a root coordinator delegating to doctor-matching, review, and scheduling agents.
 - Integrated MCP-based tool access for structured data and external actions — Cloud SQL MCP for doctor/clinic/availability lookups, Google Maps MCP for travel-time estimation, and Google Calendar MCP (via Composio) for appointment booking.
 - Implemented grounded retrieval over doctor review data using Vertex AI RAG Engine and Gemini 2.5 Flash for conversational reasoning.
 - Deployed end-to-end on Google Cloud Run.

CERTIFICATIONS

- Foundation: Introduction to LangGraph - Python | LangChain Academy | 2026 | Verify [here](#)
- Foundation: Introduction to LangChain - Python | LangChain Academy | 2026 | Verify [here](#)
- Google Cloud Gen AI Academy | Google and Hack2Skill | 2026 | Verify [here](#)
- Multi AI Agent Systems with crewAI | CrewAI | 2025 | Verify [here](#)
- Deep Learning, a 5-course specialization | [Deeplearning.ai](#) | Verify [here](#)
- MCPS: Microsoft Certified Professional, Microsoft Certification No.: 1042C4-5DHF30